

What is the vernier reading?

A 2.23 **B** 2.26 **C** 2.33 **D** 2.36

2 An aircraft, flying at a constant height, is gaining speed.

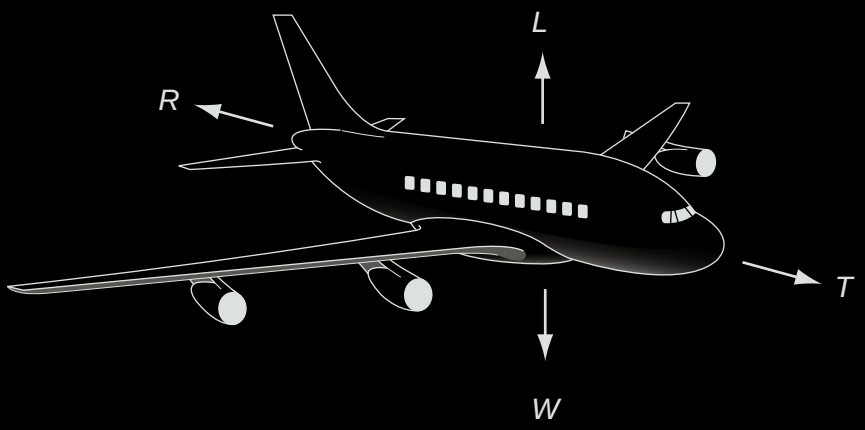
The four forces acting are

L lift due to the wings

R air resistance

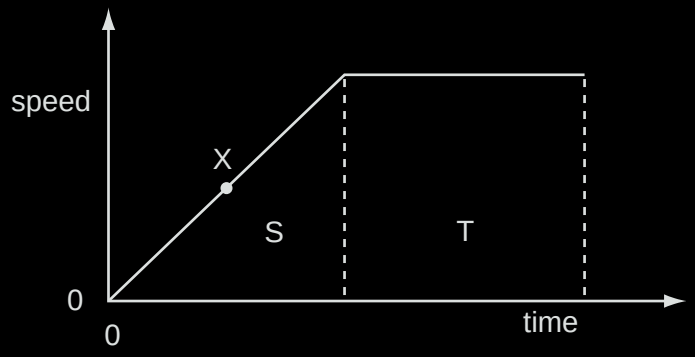
T the thrust due to the engines

W the weight



What is correct?

	vertical forces	horizontal forces
A	$L = W$	$T = R$
B	$L > W$	$T > R$
C	$L = W$	$T > R$
D	$L > W$	$T = R$



Which feature of the graph represents the distance travelled by the object whilst moving at a constant speed?

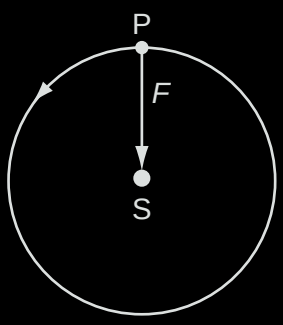
A area S

B area S + area T

C area T

D the gradient at point X

5 A body P moves in a circle around a point S. A force *F* keeps it moving in the circle.



What happens if the force *F* suddenly disappears?

A P moves directly towards S.

B P moves in a circle closer to S.

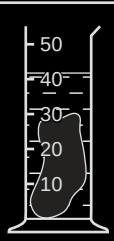
C P moves away from S in a curved path.

D P goes off in a straight line.



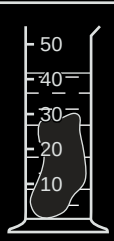
Find the mass of the stone using a mass balance

Card 1



Read the new level of the liquid in the measuring cylinder

Card 2

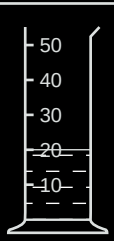


Put the stone into the liquid

Card 3

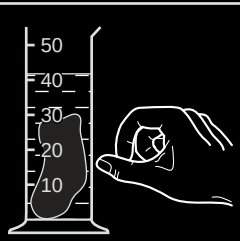
Divide the mass of the stone by its volume to find the density

Card 4



Put some liquid into a measuring cylinder and read the level of the liquid

Card 5



Flick the cylinder to get rid of any air bubbles

Card 6

Subtract the original volume of liquid from the volume of liquid plus the stone

Card 7

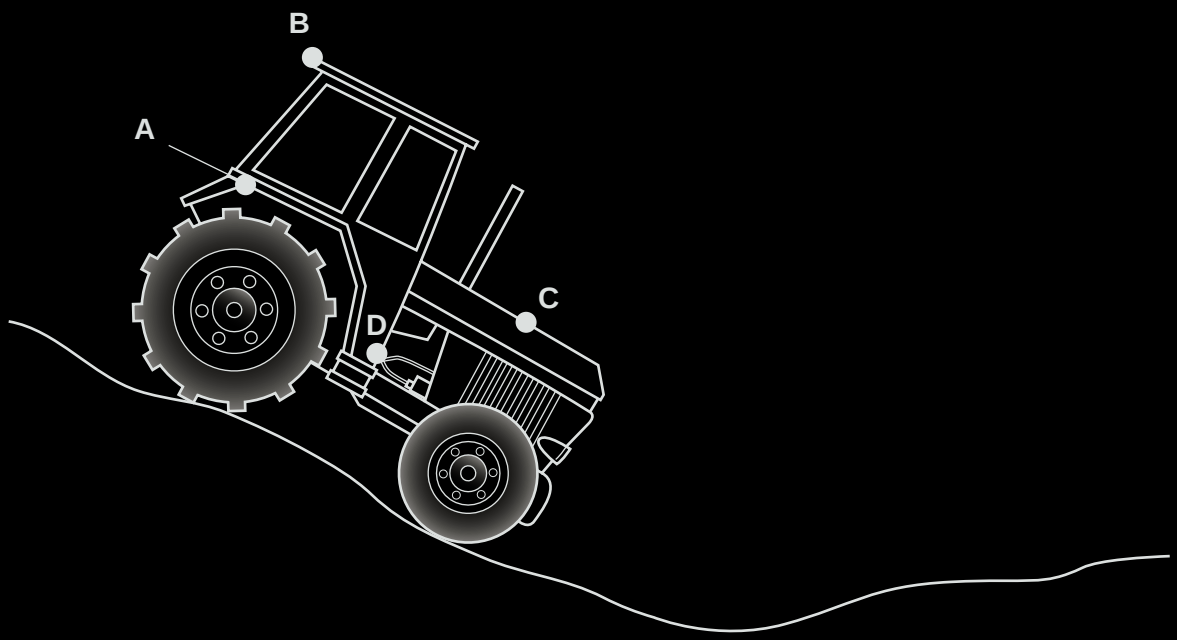
What order should the cards be in?

A $5 \rightarrow 3 \rightarrow 6 \rightarrow 2 \rightarrow 1 \rightarrow 4 \rightarrow 7$

B $1 \rightarrow 5 \rightarrow 3 \rightarrow 6 \rightarrow 2 \rightarrow 7 \rightarrow 4$

C $5 \rightarrow 6 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 7 \rightarrow 4$

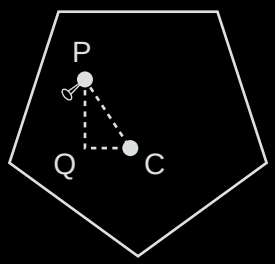
D $1 \rightarrow 4 \rightarrow 5 \rightarrow 3 \rightarrow 6 \rightarrow 2 \rightarrow 7$



9 A plane lamina is freely suspended from point P.

The weight of the lamina is 2.0 N and the centre of mass is at C.

PC = 0.50 m
PQ = 0.40 m
QC = 0.30 m



lamina

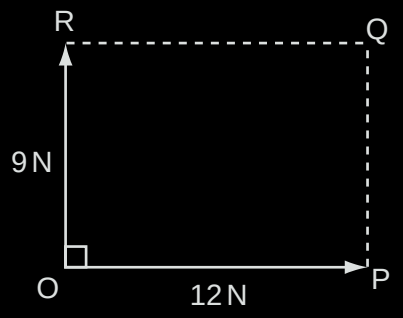
The lamina is displaced to the position shown. What is the moment that will cause the lamina to swing?

A 0.60 N m clockwise

B 0.80 N m anticlockwise

C 1.0 N m clockwise

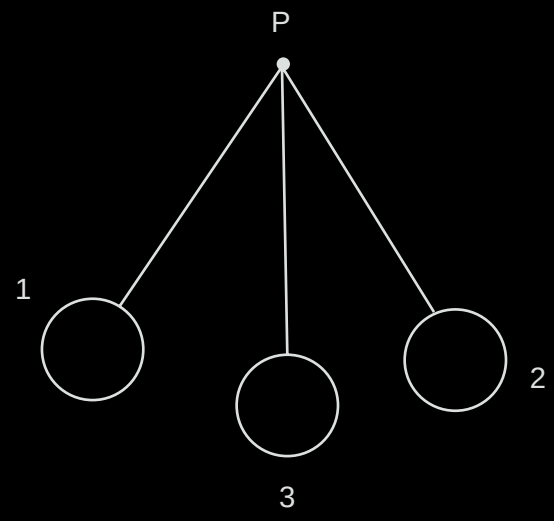
D 1.0 N m anticlockwise



What is the resultant of the forces?

	magnitude	direction
A	15 N	OQ
B	15 N	PR
C	21 N	OQ
D	21 N	PR

11 A mass hangs on a string fixed at point P. It starts from position 1 and swings to the furthest position on the opposite side, position 2. It then oscillates several times with decreasing amplitude before ending at position 3.



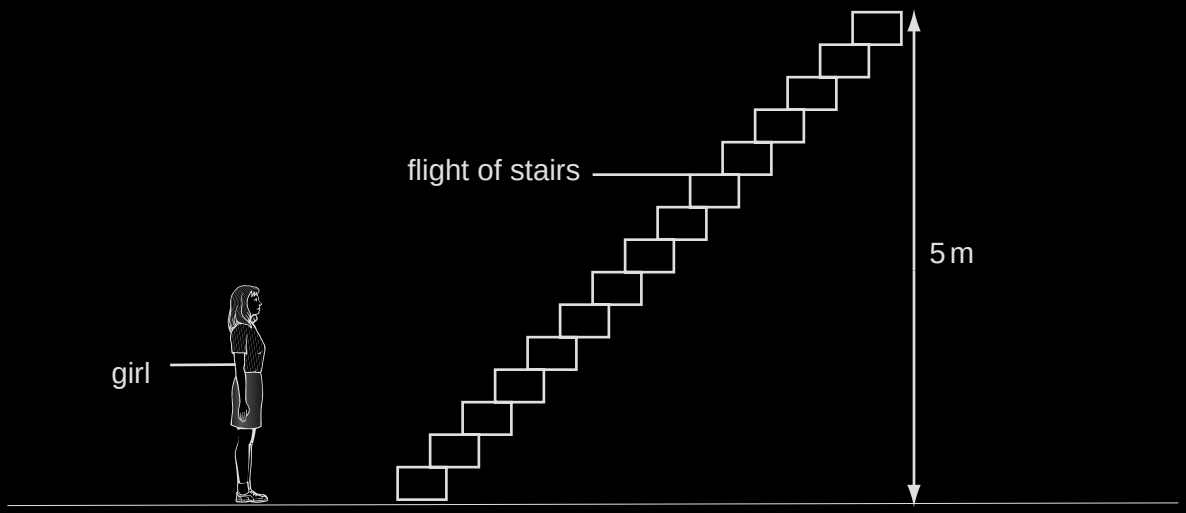
Where does the ball have the most kinetic energy?

A at position 1

B at position 2

C the first time at position 3

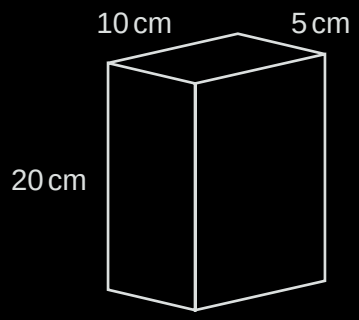
D the last time at position 3



What is the average power developed by the girl?

A 50 W **B** 100 W **C** 250 W **D** 1000 W

13 A brick of weight 80 N stands upright on the ground as shown.



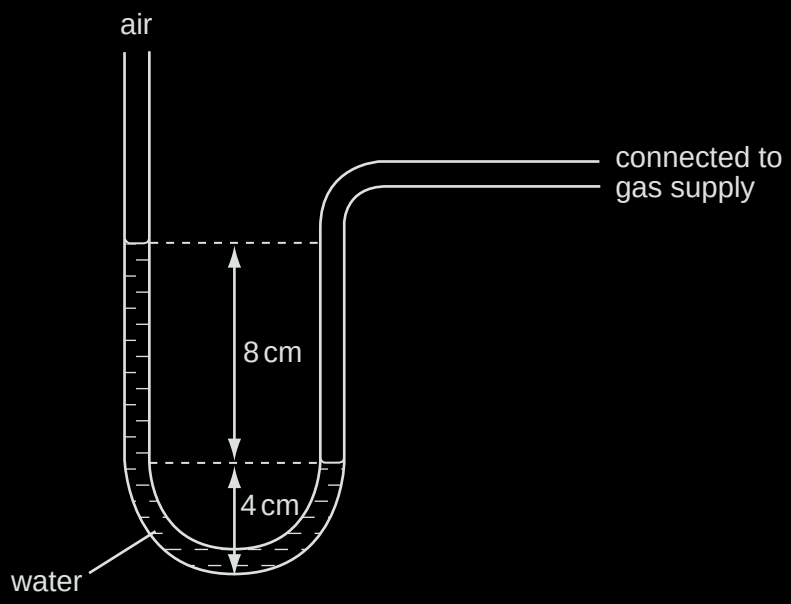
What is the pressure it exerts on the ground?

A $\frac{80}{20 \times 10}$ N / cm^2

B $\frac{20 \times 10}{80}$ N / cm^2

C $\frac{80}{10 \times 5}$ N / cm^2

D $\frac{10 \times 5}{80}$ N / cm^2



Pressure can be measured in cm of water.

What is the pressure of the gas?

A 8 cm of water more than atmospheric pressure

B 12 cm of water more than atmospheric pressure

C 8 cm of water less than atmospheric pressure

D 12 cm of water less than atmospheric pressure

15 What will **not** affect the rate of evaporation from the surface of a liquid?

A depth of the liquid

B draughts above the surface of the liquid

C surface area of the liquid

D temperature of the liquid

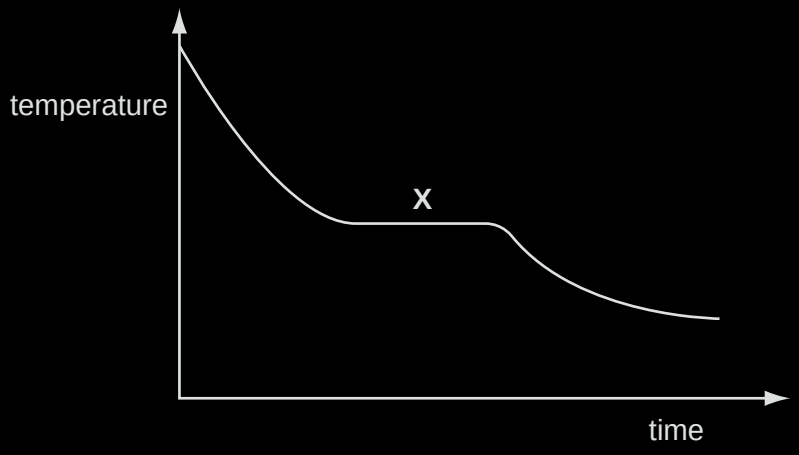
16 What makes a thermometer sensitive to small changes in temperature?

A a bulb with a thin glass wall

B a shiny liquid in its bore

C a stem with a thick glass wall

D a very narrow bore



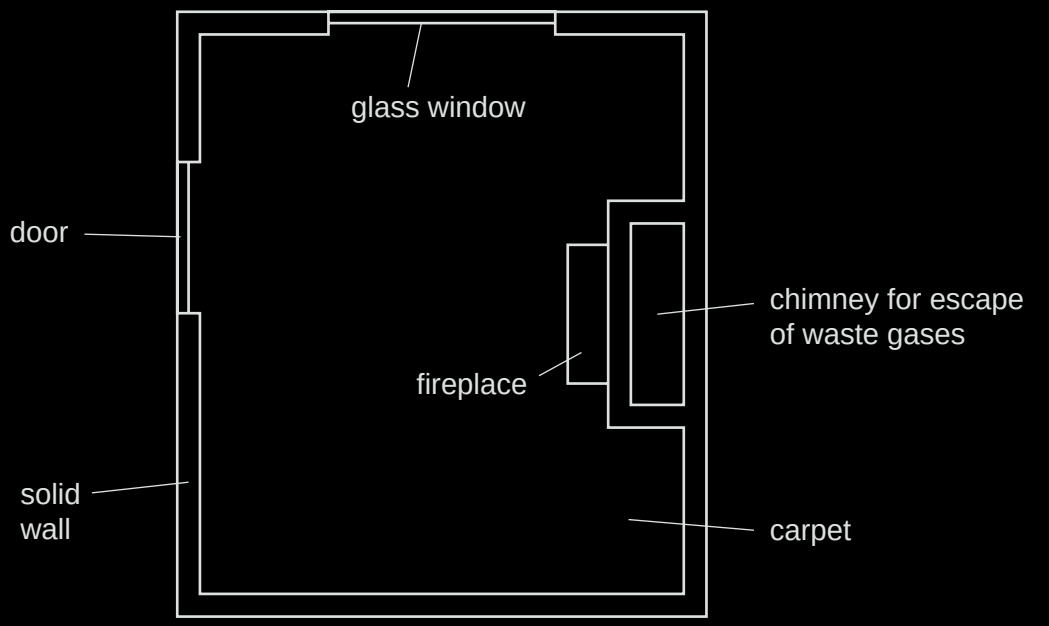
Which processes are taking place at region **X**?

A boiling and evaporation

B condensation only

C evaporation only

D solidification and evaporation



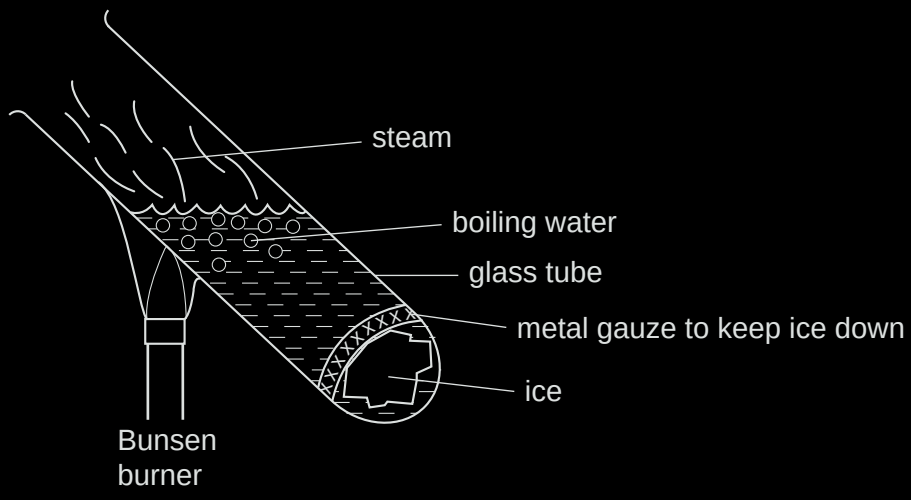
Where is most heat lost by convection?

A carpet

B chimney

C glass window

D solid wall



Why does the ice take a long time to melt, even though the water at the top of the tube is boiling?

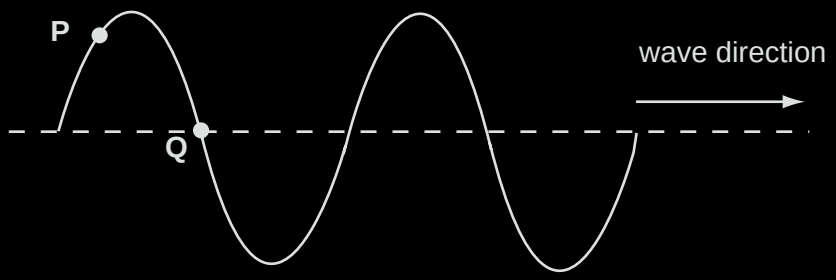
A Convection never occurs in water.

B Ice is a poor conductor of heat.

C The gauze prevents the energy reaching the ice.

D Water is a poor conductor of heat.

20 The diagram shows a wave on a string with two points **P** and **Q** marked. The wave is moving in the direction shown.



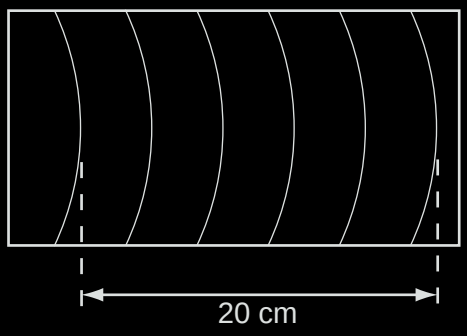
What will happen next?

A **P** will move to the right.

B **P** will move up.

C **Q** will not move.

D **Q** will move up.



What is the speed of the wave?

A 4 cm / s **B** 5 cm / s **C** 16 cm / s **D** 20 cm / s

22 Which characteristics describe an image formed in a plane mirror?

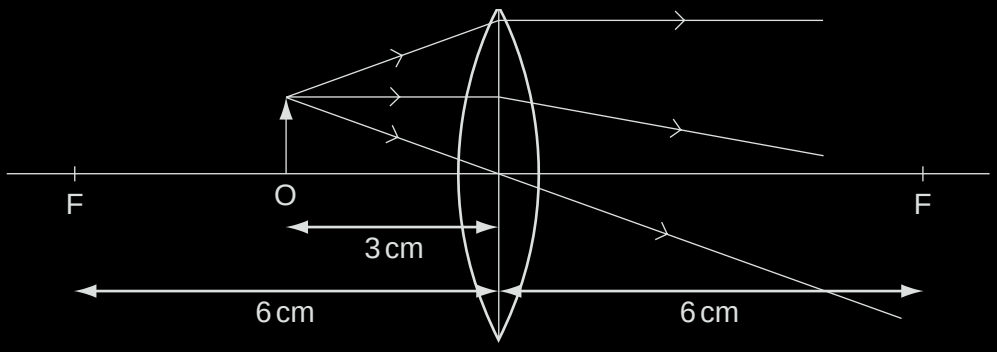
A real and inverted

B virtual and upright

C real and larger than the object

D virtual and smaller than the object

23 The diagram shows an object O placed 3 cm away from a converging lens of focal length 6 cm.



What type of image is produced?

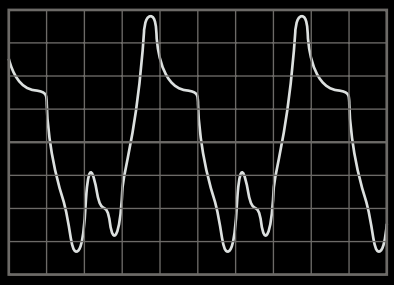
A real, erect and diminished

B real, inverted and magnified

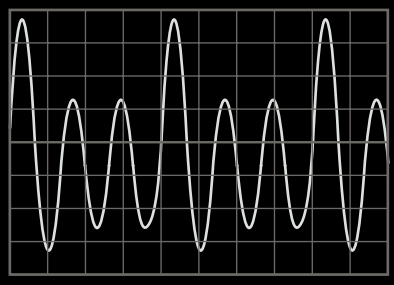
C virtual, erect and magnified

D virtual, inverted and diminished

flute



bassoon



What is the difference between the two sounds?

A the amplitude

B the frequency

C the quality (timbre)

D the wavelength

25 A student tries to magnetise a short steel rod.

Which of these tests will show that he has been successful?

A both ends of a permanent magnet attract the rod

B one end of a permanent magnet repels the rod

C the rod picks up a small piece of paper

D when freely suspended, the rod points in any direction

26 Four processes are used to charge an isolated metal sphere.

P The sphere is earthed by touching it.

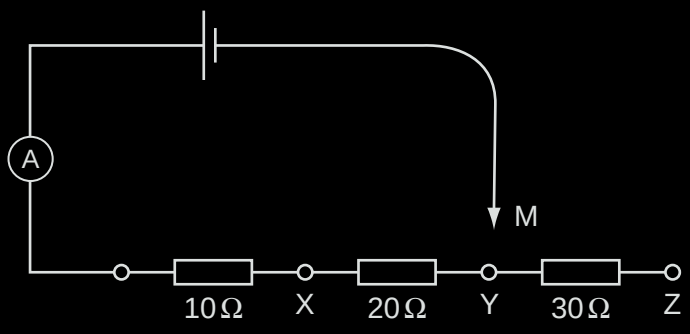
Q The earth connection is removed from the sphere.

R A charged rod is brought close to the sphere.

S The charged rod is removed.

In which order should these processes be carried out to charge the sphere?

	first ⟶ last			
A	P	Q	R	S
B	P	R	S	Q
C	R	P	Q	S
D	R	S	P	Q



When M is connected to X, the ammeter reads 0.6 A.

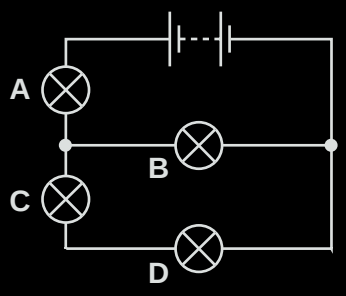
What is the ammeter reading when M is connected to Y?

A 0.1 A **B** 0.2 A **C** 0.3 A **D** 0.6 A

29 In the circuit shown, the battery lights up all four lamps.

When one of the lamp filaments melts, the other three lamps stay on.

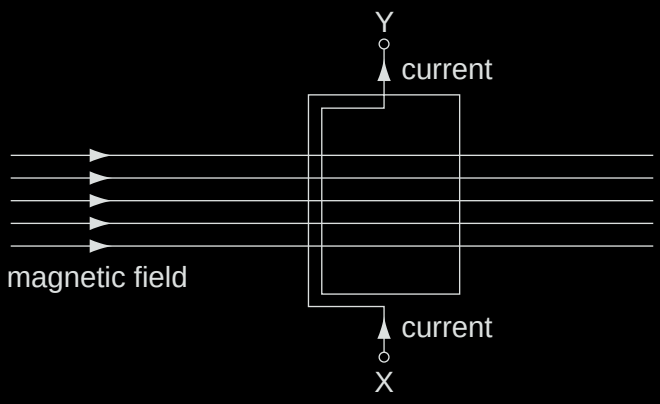
Which lamp filament melts?



30 A combined bathroom unit of a heater and a lamp is controlled by one switch. The unit contains a 2 kW heater and a 100 W lamp. In one week, the lamp uses 1 kW h of electrical energy.

How much electrical energy is used by the heater alone?

A 2 kW h **B** 4 kW h **C** 10 kW h **D** 20 kW h



In which direction does the coil move?

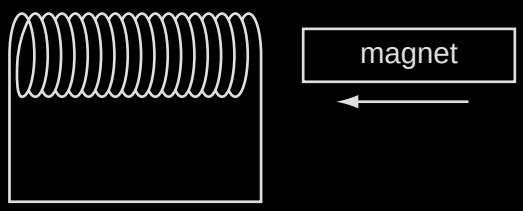
A along the magnetic field

B from X to Y

C out of the paper

D turns about the axis XY

33 A magnet is pushed horizontally towards a coil of insulated wire, inducing an e.m.f. in the coil.



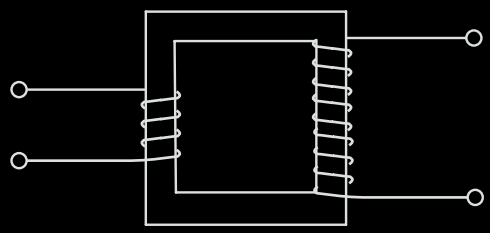
In which direction does the induced e.m.f. make the coil move?

A away from the magnet

B towards the magnet

C downwards

D upwards



Under these conditions, what output voltage and output current could be obtained?

	output voltage / V	output current / A
A	1	6
B	2	3
C	4	1
D	6	1

35 Why is a reed relay used in a switching circuit?

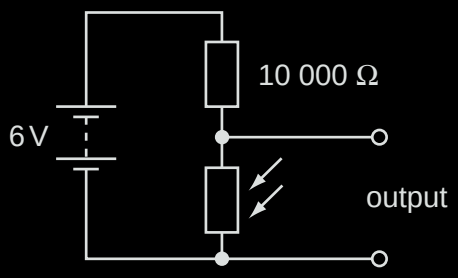
A to switch on a small current using a large current

B to switch on a small voltage using a large voltage

C to switch on a large current using a small current

D to switch on a large voltage using a large current

36 A potential divider consists of an LDR and a resistor connected to a 6 V battery.



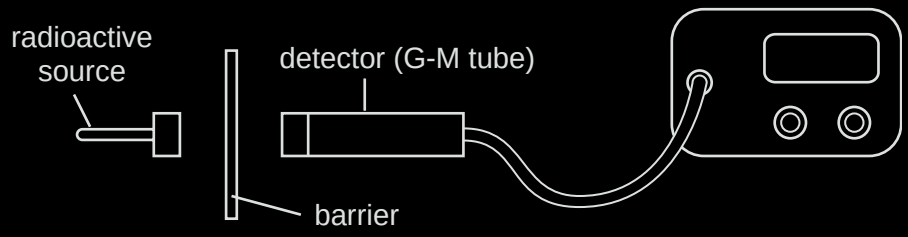
What should be the resistance of the LDR for the output to be 3 V?

A 0

B between 0 and 10 000 Ω

C 10 000 Ω

D more than 10 000 Ω



The table shows the count rates recorded by the detector for four sources.

Which source emits alpha-particles only?

source	count rate / counts per minute			
	no barrier	paper	thin aluminium	thick lead
A	200	200	200	30
B	200	30	30	30
C	1200	600	200	30
D	1200	1200	30	30

38 A nucleus is represented by $^{230}_{91}Z$. It emits one alpha-particle and then one beta-particle.

What is the resulting nucleus X?

A $^{226}_{88}X$ **B** $^{226}_{89}X$ **C** $^{226}_{90}X$ **D** $^{230}_{89}X$

39 A nuclide of the element plutonium is $^{242}_{94}Pu$.

What is the number of neutrons in its nucleus?

A 94 **B** 148 **C** 242 **D** 336

40 Which statement defines isotopes?

A two (or more) nuclides which have the same number of protons but different numbers of electrons

B two (or more) nuclides which have the same number of neutrons but different numbers of electrons

C two (or more) nuclides which have the same number of neutrons but different numbers of protons

D two (or more) nuclides which have the same number of protons but different numbers of neutrons